D2 — Attack Surface and Working Harness

1. Attack surface analysis

Channels through which content reaches the agent (Track A). One row per channel.

Channel	Agent sees it?	What is needed to inject instructions
Visible page text	Yes	Place text in the rendered page body.
Hidden DOM element (display:none)	Only with full HTML parse	Hide instructions in a non-rendered element.
HTML comment	Only if comments are kept	Put instructions in an HTML comment.
Meta description tag	Only if metadata is read	Put instructions in `<meta name="description">`.
Title tag	Yes	Put instructions in `<title>`.
Image alt attribute	Only if alt text is read	Put instructions in an `img` alt attribute.
HTTP response header	No by default	Add a custom header; needs header-aware fetching.
Tool result	Yes	Return instructions as a tool result.
Stored memory (earlier turn)	Yes	Store instructions in agent memory from a prior turn.
Email subject line	Yes	Put instructions in an email subject the agent reads.
Document metadata	No by default	Put instructions in metadata fields; needs a metadata-aware parser.
Image EXIF metadata	No by default	Put instructions in EXIF data; needs an EXIF-aware parser.

Entry points that parse untrusted input: web page HTML, tool arguments, connector responses, email and document parsers, and memory recall. Each is reachable from the agent loop once content is fetched or a tool runs. Guards that may apply: Sentinel approval for connector actions, the web-access gate for unfamiliar sites, and the per-action approval card for writes.

2. Repository

https://github.com/lordzadd/muse-research-d2

3. Exact commands to run the harness

```
git clone https://github.com/lordzadd/muse-research-d2.git
cd muse-research-d2
python3 -m pip install -r requirements.txt
python3 harness/run.py
```

Optional:

```
python3 harness/run.py --trials 5 --out results/attack_surface.csv
```

The harness prints a JSON summary with `success_rate` and writes `results/attack_surface.csv`.

AI Use Appendix

I used an AI coding assistant to draft the harness code, the attack-surface table, and this document. I reviewed and edited the output, tested the harness locally, and verified that it runs from a clean checkout with the standard library only.